Olufolajimi (Jimi) Oke

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Appointments

University of Massachusetts Amherst, MA

2019 – date *Assistant Professor*: Civil and Environmental Engineering
Director: Networks for Accessibility, Resilience & Sustainability Laboratory (NARS Lab)

Massachusetts Institute of Technology, Cambridge, MA

2016 – 2019 *Postdoctoral Associate*: Civil and Environmental Engineering (Advisor: Moshe Ben-Akiva)
Project Manager: Future Urban Mobility, MIT Energy Initiative Mobility of the Future Study

The Pennington School, Pennington, NJ

2011 – 2012 *Faculty*: Department of Mathematics; Center for Learning
2010 – 2011 *Teaching Fellow*: Center for Learning

Education

Johns Hopkins University, Baltimore, MD

2016 *Doctor of Philosophy*: Civil Engineering (Advisor: Sauleh Siddiqui)
2014 *Master of Science in Engineering*: Civil Engineering

Williams College, Williamstown, MA

2010 *Bachelor of Arts*: Physics, Music
Honors Thesis in Physics (Advisor: J. Strait); Senior Recital, Classical Guitar (Teacher: R. Phelps)

Funding

2022 – 2023 Massachusetts Department of Transportation ($125,000), *Robust decision-making framework for sustainable operations and planning of MBTA rapid transit vehicles*, **PI**; Co-PIs: Eleni Christofa and Eric Gonzalez

2021 – 2023 Pioneer Valley Transit Authority/US DOT ($106,000) *Pioneer Valley Transit Review and Improvement Planning Study (PV-TRIPS)*, PI: Camille Barchers; **Co-PI**

2021 – 2022 American Public Power Association ($191,197), *Cost-benefit analysis of co-deploying optical fiber and electric cabling underground*, **PIs**: Anna Goldstein, Jimi Oke; Co-PIs: Jay Taneja, Michael Bloomberg

2021 – 2022 Pioneer Valley Transit Authority ($73,952), *Installation of On-Bus Mobile Ticket Validators and Development of an Innovative Origin-Destination-Transfer (ODX) Model*, **PI**

2020 National Science Foundation (UMass Share: $352,355), *America's Water Risk: Water System Data Pooling for climate vulnerability assessment and warning system*, UMass PI: Casey Brown; **Senior Personnel**

2020 – 2021 Massachusetts Department of Transportation ($122,522), *Tracking and Reducing the Energy, Emissions and Costs of Rapid Transit Vehicles (TREEM)*, **PI**; Co-PIs: Eleni Christofa and Eric Gonzalez

Awards & Fellowships

University of Massachusetts Amherst

2021 Brack Endowment Award for Research, *Department of Civil and Environmental Engineering*

Massachusetts Institute of Technology

2019 Kaufman Teaching Certificate, *Teaching & Learning Lab*
2017 Postdoctoral Teaching Fellowship, *Department of Civil and Environmental Engineering*
2013 Half-tuition Scholarship: "Modeling and Simulation of Transportation Networks", *MIT Professional Education Short Programs*

Johns Hopkins University

2015 Gordon Croft Fellowship, *Environment, Energy, Sustainability & Health Institute (E^2SHI)*
2015 Article selected for Promotion, Elsevier, *Journal article "Tracking global bicycle ownership patterns"*
2015 Civil Engineering Graduate Service Award, *G.W.C. Whiting School of Engineering*
2015 Teaching-as-Research Fellowship, *Center for the Integration of Research, Teaching and Learning*
2013 Educational Training Core Traineeship, *Global Obesity Prevention Center*
2012 Whiting School of Engineering Research Fellowship

Williams College

2010 Howard P. Stabler Prize in Physics
2010 William W. Kleinhandler Prize for Excellence in Music
2010 Sigma Xi Honors
2008, 09 Williams College Summer Science Research Fellowship

Publications

Journal Articles

[J23] M. Mohammed, **J. Oke**, *A spatiotemporal trip chaining framework for a regional transit system with open mobile automated fare collection*, (*In preparation*).

[J22] Z. Han, E. Gonzales, E. Christofa, **J. Oke**, *Line-specific energy modeling framework for urban rail transit systems: A case study of Boston*, (*Under review*).

[J21] M. Arabi, **J. Oke**, *System-wide energy modeling of bus transit systems: A case study of the Pioneer Valley*, (*Under review*).

[J20] M. Mohammed, **J. Oke**, *Efficient extraction of spatiotemporal bus passenger trip patterns using time series alignment of partial mobile ticketing data*, (*Under review*).

[J19] N. Apostolov, **J. Oke**, R. Suttle, S. Arwade, B. Kane, *Predicting tree failure likelihood for utility risk mitigation via artificial intelligence*, (*Under review*).

[J18] H. Chung, E. Kumpel, **J. Oke**, *Drinking water accessibility typologies in low- and middle-income countries*, (*Under review*).

[J17] M. Arabi, S. Gerasimidis, C. Barchers, **J. Oke**, *Resilience in multilayer mobility infrastructure networks: a review and conceptual framework for equity-based assessment*, (*Under review*).

[J16] S. Goodarzi, H. F. Kashani, **J. Oke**, C. L. Ho, *Stochastic analysis for estimating track geometry degradation rates based on GPR and LiDAR data*, (*Under review*).

[J15] M. Mohammed, **J. Oke**, *Origin-destination inference in public transportation systems: a comprehensive review*, International Journal of Transportation Science and Technology (*In press*).

[J14] Z. Han, E. Gonzales, E. Christofa, **J. Oke**, *Modeling system-wide urban rail transit energy consumption: A case study of Boston*, Transportation Research Record, 2676(12): 627-640 (2022).

[J13] S. Goodarzi, H. F. Kashani, **J. Oke**, C. L. Ho, *Data-driven methods to predict track degradation: a case study*, Construction and Building Materials, 344: 128166 (2022).

[J12] N. Kumar, **J. Oke**, B. Nahmias-Biran, *Activity-based epidemic propagation and contact network scaling in auto-dependent metropolitan areas*, Scientific Reports, 11(1): 1-14 (2021).

[J11] B. Nahmias-Biran, **J. B. Oke**, N. Kumar, *Who Benefits from AVs? Social Implications of Autonomous Vehicle Policies in Full-Scale Cities*, Transportation Research Part A, 154: 92-107 (2021).

[J10] B. Nahmias-Biran, **J. B. Oke**, N. Kumar, C. L. Azevedo, M. Ben-Akiva, *Evaluating the impacts of shared automated mobility on-demand: an activity-based accessibility approach*, Transportation, 48: 1613–1638 (2021).

[J9] **J. B. Oke**, A. P. Akkinepally, S. Chen, Y. Xie, Y. M. Aboutaleb, C. L. Azevedo, C. Zegras, J. Ferreira, M. Ben-Akiva, *Evaluating systemic effects of automated on-demand services through large-scale, agent-based simulation of auto-dependent, prototype cities*, Transportation Research Part A, 140: 98-126 (2020).

[J8] D. A. Martinez, J. Cai, **J. B. Oke**, A. Jarrell, F. Feijoo, J. Appelbaum, E. Klein, S. Barnes, S. R. Levin, *Where is my Infusion Pump? Harnessing Network Dynamics for Improved Hospital Equipment Fleet Management*, Journal of the American Medical Informatics Association, 27(6): 884-892 (2020).

[J7] **J. B. Oke**, Y. M. Aboutaleb, C. L. Azevedo, Y. Han, A. Akkinepally, P. C. Zegras, J. Ferreira, M. E. Ben-Akiva, *A novel global urban typology framework for sustainable mobility futures*, Environmental Research Letters, 14(9): 95006 (2019).

[J6] B. Nahmias-Biran, **J. B. Oke**, C. L. Azevedo, N. Kumar, A. Araldo, K. Basak, R. Seshadri, M. Ben-Akiva, *From traditional to automated mobility on demand: a comprehensive framework for modeling mobility on demand services in SimMobility*, Transportation Research Record, 2673(12): 15-29 (2019).

[J5] **O. Oke**, D. Huppmann, M. Marshall, R. Poulton, S. Siddiqui, *Multimodal transportation flows in energy networks with an application to crude oil markets*, Networks and Spatial Economics, 19(2): 521-555 (2019).

[J4] **O. Oke**, K. Bhalla, D. C. Love, S. Siddiqui, *Spatial associations in global bicycle ownership*, Annals of Operations Research, 263: 529-549 (2018).

[J3] **O. Oke**, K. Bhalla, D. C. Love, S. Siddiqui, *Tracking global bicycle ownership patterns*, Journal of Transport and Health, 2(4): 490-501 (2015).

[J2] **O. Oke**, S. Siddiqui, *Efficient automated schematic map drawing using multiobjective mixed integer programming*, Computers and Operations Research, 61:1-17 (2015).

[J1] C. Chudzicki, **O. Oke**, W. K. Wootters, *Entanglement and Composite Bosons*, Physical Review Letters, 104(7):070402 (2010).

Conference Papers & Extended Abstracts [Peer Reviewed]

[C11] Z. Han, E. Gonzales, E. Christofa, **J. Oke**, *Line-specific energy modeling framework for urban rail transit systems: A case study of Boston*, TRB Annual Meeting (2022).

[C10] M. Arabi, **J. Oke**, *System-wide Energy Modeling of Bus Transit Systems: A Case Study of The Pioneer Valley*, TRB Annual Meeting (2022).

[C9] M. Mohammed, **J. Oke**, *Spatiotemporal Trip Chaining Framework for Open Mobile Fare Collection Systems*, TRB Annual Meeting (2022).

[C8] N. Apostolov, **J. Oke**, *Retrospective Investigation of Country Activity and Mobility Patterns and Their Interdependencies on Early-onset COVID-19 Outcomes*, TRB Annual Meeting (2022).

[C7] S. Goodarzi, H. F. Kashani, **J. Oke**, C. L. Ho, *Stochastic Analysis for Estimating Track Geometry Degradation Rates Based on GPR and LiDAR Data*, TRB Annual Meeting (2022).

[C6] Z. Han, E. Gonzales, E. Christofa, **J. Oke**, *Modeling system-wide urban rail transit energy consumption: A case study of Boston*, TRB Annual Meeting (2022).

[C5] M. Mohammed, **J. Oke**, *Trip pattern typologies in the Pioneer Valley bus transit system*, TRB Annual Meeting (2022).

[C4] B. Nahmias-Biran, **J. B. Oke**, N. Kumar, *Activity-based contact network scaling and epidemic propagation in metropolitan areas*, TRB Annual Meeting (2021).

[C3] B. Nahmias-Biran, **J. B. Oke**, N. Kumar, A. P. Akkinepally, C. L. Azevedo, P. C. Zegras, J. Ferreira, M. Ben-Akiva, *Who Benefits from AVs? Equity Aspects of Autonomous Vehicles Policies in a Full-Scale Prototype Cities*, TRB Annual Meeting (2020).

[C2] E. Gross, **J. Oke**, A. P. Akkinepally, B. Nahmias-Biran, C. L. Azevedo, C. Zegras, J. Ferreira, M. Ben-Akiva, *Accessibility and energy consumption evaluation under different strategies of mobility on-demand deployment*, TRB Annual Meeting (2019).

[C1] Y. Han, **J. Oke**, S. Hua, J. Zhou, C. L. Azevedo, C. Zegras, J. Ferreira, M. Ben-Akiva, *Global urban typology discovery with a latent class choice model*, TRB Annual Meeting (2018).

Theses and Reports

[R5] W. H. Green et al., *Insights Into Future Mobility: A Report from the Mobility of the Future Study*, MIT Energy Initiative, Cambridge, MA, November 2019.

[R4] **O. Oke**, *Network modeling and optimization for energy and sustainable transit*, Doctoral Dissertation in Civil Engineering, Johns Hopkins University, May 2016.

[R3] **J. Oke**, S. Siddiqui, K. Bhalla, D.C. Love, J. De Vito, M. Van Doren, Max Marshall, *Making Baltimore More Bike Friendly*, Department of Civil Engineering, Johns Hopkins University, March 2014.

[R2] **O. Oke**, *Bicycling in Baltimore: key concerns*, Report (submitted to Baltimore DOT), Department of Civil Engineering, Johns Hopkins University, January 2013.

[R1] **O. Oke**, *A nonlinear optical loop mirror modelocked fiber laser*, Honors Thesis in Physics, Williams College, May 2010.

Communications

Invited Presentations

[IP16] *Mobility and epidemics in urban typologies*, Diversity, Equity and Inclusion Seminar, Department of Civil and Environmental Engineering, Massachusetts Institute of Technology, MA, June 2020.

[IP15] *Analysis of future mobility on-demand systems in global urban typologies*, INFORMS Annual Meeting, Seattle, WA, October 2019.

[IP14] *New urban typologies for sustainable mobility*, Department of Civil and Environmental Engineering, University of Illinois Urbana-Champaign, IL, February 2019.

[IP13] *New urban typologies for sustainable mobility*, Department of Civil and Environmental Engineering, University of Massachusetts Amherst, MA, February 2019.

[IP12] *Discovering new urban typologies*, Department of Civil and Environmental Engineering, University of Pittsburgh, PA, January 2019.

[IP11] *Discovering sustainable future urban mobility policies via simulation of prototype cities from global urban typologies*, Systematizing and upscaling urban solutions for climate change mitigation, Berlin, Germany, September 2018.

[IP10] *Discovering Robust Urban Mobility Futures via Agent Based Simulation in Prototype Cities*, 7th TRB Innovations in Travel Modeling Conference, Atlanta, GA, June 2018.

[IP9] *Need for and Uses of Risk Analysis: Technical approaches from the university perspective*, Risk Analysis Workshop, 7th TRB Innovations in Travel Modeling Conference, Atlanta, GA, June 2018.

[IP8] *Urban mobility simulation for scenario discovery in globally representative prototypes*, Physics Colloquium, Department of Physics, Williams College, MA, October 2017.

[IP7] *Exploring sustainable mobility strategies in future cities*, Log Lunch Series, Center for Environmental Studies, Williams College, MA, October 2017.

[IP6] *A Crude Oil Market Model for the United States*, INFORMS Annual Meeting, Philadelphia, PA, November 2015.

[IP5] *An Equilibrium Model of the US Crude Oil Market*, the 22nd International Symposium on Mathematical Programming, Pittsburgh, PA, July 2015.

[IP4] *Redefining Infrastructural Space*, Environment, Energy, Sustainability & Health Institute Art/Science Roundtable, Maryland Institute College of Art, Baltimore, May 2015.

[IP3] *Tracking Global Bicycle Ownership Patterns*, INFORMS Data Mining and Analytics Workshop, San Francisco, CA, November 2014.

[IP2] *An efficient automated multiobjective programming approach to map schematization*, Systems/Policy/Energy Seminar, Johns Hopkins University, March 2014.

[IP1] *A Mixed-integer Programming Tool for Creating Effective Schematic Urban Transit Maps*, INFORMS Annual Meeting, Minneapolis, MN, October 2013.

Contributed Presentations

[CP12] *Visual Interpretation of Convolutional Neural Networks for Tree-risk Prediction and Conflict Detection*, INFORMS Annual Meeting, Indianapolis, IN, October 2022.

[CP11] *Digging into the Data: Cost-benefit Analyses of Undergrounding Strategies in Electric and Broadband Networks*, INFORMS Annual Meeting, Indianapolis, IN, October 2022.

[CP10] *Spatio-temporal Trip Pattern Typology Analysis for a Regional Bus Network*, INFORMS Annual Meeting, Indianapolis, IN, October 2022.

[CP9] *Framework for Line-specific Urban Rail Transit Energy Prediction and Inference*, INFORMS Annual Meeting, Indianapolis, IN, October 2022.

[CP8] *Assessing the energy impacts of automated on-demand service deployment strategies in sprawling, auto-dependent cities*, Transatlantic Infraday Conference, Federal Energy Research Commission, Washington, DC, November 2018.

[CP7] *Discovering Urban Typologies For Future Mobility Scenarios In Prototype Cities*, INFORMS Annual Meeting, Houston, TX, October 2017.

[CP6] *Analyzing United States Crude Oil Flows*, Transatlantic Infraday Conference, Federal Energy Research Commission, Washington, DC, October 2015.

[CP5] *An Oil Market Model for the United States*, Modeling and Optimization: Theory and Applications, Lehigh University Department of Industrial and Systems Engineering, Bethlehem, PA, July 2015.

[CP4] *Solving the Crude on Rail Problem using an Equilibrium Model of the US Crude Oil Market*, Critical Infrastructure Symposium, Linthicum, MD, April 2015.

[CP3] *Global Bicycle Availability*, INFORMS Annual Meeting, San Francisco, CA, November 2014.

[CP2] *Multiobjective optimization for automatic schematic map drawing*, Modeling and Optimization: Theory and Applications, Lehigh University Department of Industrial and Systems Engineering, Bethlehem, PA, August 2014.

[CP1] *Schematic map automation and optimization*, Civil Engineering Graduate Seminar, Johns Hopkins University, November 2013.

Teaching

University of Massachusetts Amherst, MA

Instructor, Department of Civil and Environmental Engineering

Spring 2023 Probabilistic Machine Learning (Graduate)

Spring 2022 Big Data and Machine Learning for Engineers (Graduate)
15 students

Fall 2021 Probability & Statistics in Civil Engineering (Undergraduate)
116 students

Spring 2021 Big Data and Machine Learning for Engineers (Graduate)
9 students

Fall 2020 Probability & Statistics in Civil Engineering (Undergraduate)
60 students

Spring 2020 Data Mining and Machine Learning for Engineers (Graduate)
16 students; created new course

Fall 2019 Probability & Statistics in Civil Engineering (Undergraduate)
127 students; introduced MATLAB programming assignments

Massachusetts Institute of Technology, MA

Teaching Assistant, Department of Civil and Environmental Engineering

Fall 2018 Transportation Systems Analysis: Demand and Economics (Graduate)
15 students, recitations, content development

Summer 2018 Modeling and Simulation of Transportation Networks (Professional)
content development

Fall 2017 Transportation Systems Analysis: Demand and Economics (Graduate)
25 students, recitations, content development, assessments

Teaching Fellow, Department of Civil and Environmental Engineering

Spring 2016 Multivariate Data Analysis (Undergraduate)

16 students, recitations, lectures, Jupyter notebook development, innovative assessments, project

Johns Hopkins University, MD

Instructor, Hopkins Engineering Applications & Research Tutorials Program

Fall 2015 Reality Distortion: The Impact and Automation of Schematic Maps (Undergraduate)

3 students, content development: GAMS, Python, optimization, visualization

Teaching-as-Research Fellow, Department of Civil Engineering

Spring 2015 Probability and Statistics in Civil Engineering (Undergraduate)

97 students, designed 12 applied MATLAB projects, surveys & focus group to measure impact

Teaching Assistant, Department of Civil Engineering

Oct 2013 Optimization and Equilibrium Modeling in Systems Engineering (Graduate, day-long)

11 students, GAMS installation and programming help

Spring 2013 Probability and Statistics in Civil Engineering (Undergraduate)

55 students, office hours, wrote and graded quizzes & exam problems

Fall 2012 Statics and Mechanics of Materials (Undergraduate)

110 students, taught 10 of 12 lab sections, assessed lab reports, course grade manager

The Pennington School, NJ

2011 – 2012 **Faculty Member**, Mathematics Department

Algebra II Honors, Precalculus (Course Leader)

developed innovative assessments, redesigned Precalculus curriculum

2010 – 2012 **Teaching Fellow – Faculty Member**, Center for Learning

Algebra II, Math Skills Tutorial, Writing Skills Tutorial, Communication Skills

small classes, students with learning differences, focused mentorship

Williams College, MA

Physics Tutor, Office of Academic Resources

2008 – 2010 Peer Tutor Program

one-on-one appointments several times a week

2007 – 2010 Math and Science Resource Center

walk-in/group sessions, 2-hour shifts, 2 – 4 times a week

Teaching Assistant, Department of Physics

Spring 2009 "Waves and Optics"

grading, homework assistance

Fall 2008 "Particles and Waves, Enriched"

grading, homework assistance

Teaching Assistant, Department of Art

Fall 2007 Drawing

studio manager, curated student work, occasional modeling

Service

Graduate Research Mentorship

2021–date	Mahsa Arabi (PhD[1], UMass Amherst)
2021–date	Mohammed Mohammed (MSCE[1]/PhD, UMass Amherst)
2020–date	Atanas Apostolov (MSCE[1], UMass Amherst)
2020–date	Zhuo Han (MSCE[1]/PhD, UMass Amherst)
2018–19	Yifei Xie (MST[2], MIT)
2018–19	Siyu Chen (MST[2], MIT)
2017–18	Eytan Gross (MST[2], MIT)
2017–19	Youssef Medhat (MST[2], MIT)
2017–18	Iveel Tsogsuren (MEng, MIT)
Summer 2017	Jin (Jasmine) Zhou (MA, Columbia University)[3]
Spring 2017	Scott Foster (Leaders for Global Operations Fellow, MIT)
Fall 2016	Akshay Padmanabha (MEng, MIT)
2016–17	Sean Hua (MEng, MIT)
2016–17	Michael Choi (MEng, MIT)

Undergraduate Research Mentorship

2022-date	Alexa Weinman (BSE 2023, Civil and Environmental Engineering, UMass Amherst)
2020-2021	Hichul Chung (BSE 2021, Mechanical Engineering, UMass Amherst)
Summer 2020	Adegbola Fayemi (College of Wooster)
2017–18	Sharlene Chiu (Super UROP[4], MIT)
Spring 2017	Joseph Noszek (UROP[4], MIT)
Spring 2017	Abenezer Samuel (UROP[4], MIT)
Winter 2017	Gabriel Madonna (Mini UROP[4], MIT)
2013–16	Max Marshall (JHU)
2014–15	Ricky Poulton (JHU)
2013–14	Molly Van Doren (JHU)

Internal Service, Department and School/College

University of Massachusetts Amherst

2021–date	College of Engineering ECS-IT Advisory Committee Member (CEE representative)
2021–22	Thesis Committee Member: Brent Scott (PhD, 2022)
2021	Thesis Committee Member: Peetak Mitra (PhD, 2021)
2019–20	Thesis Committee Member: Haralampos Sipetas (PhD, 2020)

Civil & Environmental Engineering Postdoc Committee, Massachusetts Institute of Technology

2016–17	Member

Homewood Gradaute Board, Johns Hopkins University

2015–16	Graduate Representative, Whiting School of Engineering

[1] Masters of Science in Civil Engineering
[2] Masters of Science in Transportation
[3] Now PhD student at University of Southern California
[4] Undergraduate Research Opportunities Program

Civil Engineering Graduate Association, Johns Hopkins University

2015–16 President

2014–15 Social Coordinator (Founder)

Graduate Seminars, Department of Civil Engineering, Johns Hopkins University

2014–15 Cochair (Systems track)

2013–14 Committee member

External Advising

2022–2024 Secondary Advisor and Thesis Committee Member: Nishant Kumar, PhD candidate (ETH Zurich)

Manuscript Review

2023 Transportation Research: Part A

2022 Transportation Research: Part A, Part C, Part D; Transportation Research Board; Energy Journal

2021 Energy Research & Social Science; Transportation Research: Part C; Journal of Urban Planning and Development

2020 Transportation Research Record

2019 Transportation Research Board

2018 Transportation Research: Part A; International Journal of Geographical Information Science; Journal of Transport & Health

2017 International Journal of Sustainable Transportation; Computational Optimization and Applications

2016 Optimization and Engineering

2015 IEEE Transactions on Power Systems

2014 INFORMS Data Mining & Analytics Workshop

Conference Organization

2023 Session Chair, INFORMS Annual Meeting

2017 Session Chair, INFORMS Annual Meeting

Outreach

2014 – 16 Graduate Student Mentor, STEM Achievement in Baltimore Elementary Schools (SABES)

2015 Contributor, Baltimore City Bicycle Master Plan 2015

2013 Project co-coordinator, "Making Baltimore More Bicycle Friendly"

Academic Affiliations

American Society of Civil Engineers (ASCE)

Association for Computing Machinery

(Special Interest Group on Knowledge Discovery and Data Mining)

Institute for Operations Research and Management Sciences (INFORMS)

Network Science Society

Transportation Research Board